# Divya Makkenchery

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## EDUCATION

**The University of Chicago** Chicago, IL
M.S. in Computer Science, Specialization in Artificial Intelligence March 2026
Coursework: Introduction to Cryptography, UNIX Systems, Machine Learning, Databases, Generative AI

**George Washington University** Washington, DC
B.S. in Cognitive Neuroscience, Minors in Mathematics & Business May 2023
Coursework: Cognitive Psychology, Human Neurobiology, Data Science and Artificial Intelligence
**Honors***:* Presidential Scholarship Recipient

## SKILLS

**Technical Languages:** Python, C, LaTeX, SQL, R, MATLAB
**Data & ML Tools:** Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, MySQL, Jupyter Notebook
**Generative AI:** Retrieval-Augmented Generation (RAG), LLM Applications, Prompt Engineering, AI Agents, Vector Databases, Embedding Models, LangChain, OpenAI API
**Techniques:** Data Cleaning, Data Validation, Statistical Analysis, Data Visualization
**Software:** Excel, PowerPoint, Word, SharePoint, Adobe Premiere, REDCap
**Extracurriculars:** Bhangra Blowout Co-Director responsible for event planning and $50,000+ budget management

## PROFESSIONAL EXPERIENCE

**Los Alamos National Laboratory** Los Alamos, NM
*Data Science Intern* June 2025-August 2025

- Authored 60+ page stakeholder reports translating statistical trends and community-level indicators into actionable policy recommendations for local nonprofits and the state government of New Mexico
- Engineered ETL-style data processing pipelines in Python (Pandas) for 100+ heterogeneous nonprofit datasets, cleaning and transforming inconsistent schemas, missing values, duplicate records, and conflicting measurement units to support downstream analysis.
- Built validation and normalization workflows using cross-dataset consistency checks and duplicate detection, improving dataset quality by up to 90% across datasets ranging from hundreds to thousands of records.
- Debugged and optimized 20+ Python and R scripts by tracing runtime failures, identifying logic errors in data transformations, and validating outputs against stakeholder requirements for a 17-person analytics team.

**GW Autism and Neurodevelopmental Disorders Institute** Washington, DC
*Research Intern* January 2023-May 2023

- Collaborated on survey design and data collection via REDCap, for various research studies on the development of social decision making in children with autism.
- Developed MATLAB scripts to preprocess experimental stimuli datasets by detecting duplicate image entries and generating randomized, non-overlapping image sets for controlled behavioral studies.
- Designed a randomized sampling workflow in MATLAB to automate stimulus selection while preserving non-overlapping image distributions across experimental groups.

## PROJECT EXPERIENCE

**Refactor 312, CARE Non-Profit |** Python, Pandas, Matplotlib June 2026 - August 2026

- Analyzed volunteer engagement and retention data using Python, Pandas, and Matplotlib, examining participation frequency, tenure, demographics, and professional sectors.
- Applied K-Means clustering and the elbow method to segment volunteers into three groups and identify trends in volunteer behavior and retention.
- Conducted exploratory analysis of student and school data to assess program impact.

**E2E Database Design & Query System |** MySQL, Python, SQL October 2025 - December 2025

- Engineered a full relational database system by defining requirements, producing a normalized ER diagram and relational schema aligned with functional dependencies.
- Implemented the database in MySQL, inserting real datasets via Python; developed GUI front-end that executes predefined SQL queries, supports user inputs, and allows insertion of new records through a structured application interface.